TECHNICAL PROJECT PRESENTATION & APPLICATION WALKTHROUGH

DevOps Incident Watchdog: Autonomous SRE Telemetry & Closed-Loop Sprint Backlog Automation

Monitored Microservice: FlashMart Tech (10-Minute E-Commerce at VIT Vellore Hub)

Submitted by:

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Course: ISWE406 Agile Development Process and DevOps

School of Computer Science Engineering and Information Systems (SCOPE)
Vellore Institute of Technology, Vellore Campus | Academic Year 2026-2027

Executive Problem Statement & Project Scope

Why Traditional DevOps Fails During Promotional Flash Spikes:

1. Extreme Concurrency & Dark Store Exhaustion: In quick-commerce platforms (FlashMart, Zepto, Blinkit), promotional flash sales generate sudden 50x demand spikes that deplete localized warehouse stock to zero within seconds.

2. Unhandled Microservice Boundary Crashes: Algorithms computing discount allocations or delivery slots encounter fatal ZeroDivisionError when stock drops to 0, crashing WSGI worker threads with HTTP 500.

3. Crippling Administrative Delay (Elevated MTTA): Traditional monitoring leaves crashes in raw CloudWatch streams. Engineers spend 2 to 8 hours manually locating logs, reproducing errors, and typing Jira/GitHub tickets.

4. The DevOps Incident Watchdog Solution: An event-driven, closed-loop SRE platform that autonomously captures crashes in CloudWatch, archives forensic JSON dumps in Amazon S3, mints structured Agile Bug Stories on GitHub, and enables automated CI/CD closure upon verified fix.

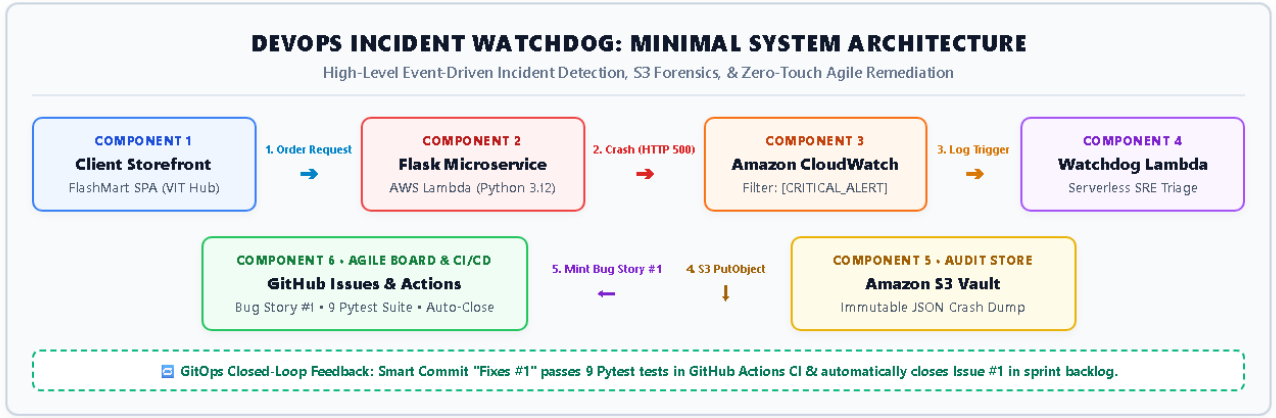
System Architecture: Decoupled Three-Tier Topology

Core Subsystems & Responsibilities:

Presentation Tier: FlashMart Single-Page Application (SPA) with dynamic delivery hub routing defaulting to VIT Vellore Campus.

Serverless Compute Tier: AWS Lambda microservices (Python 3.12, Flask adapter) coupled with asynchronous Watchdog triage worker.

Cloud & Agile Tier: Amazon CloudWatch real-time metric ingestion, Amazon S3 immutable forensic vault, and GitHub Issues Kanban board.



MICROSERVICE FEATURES

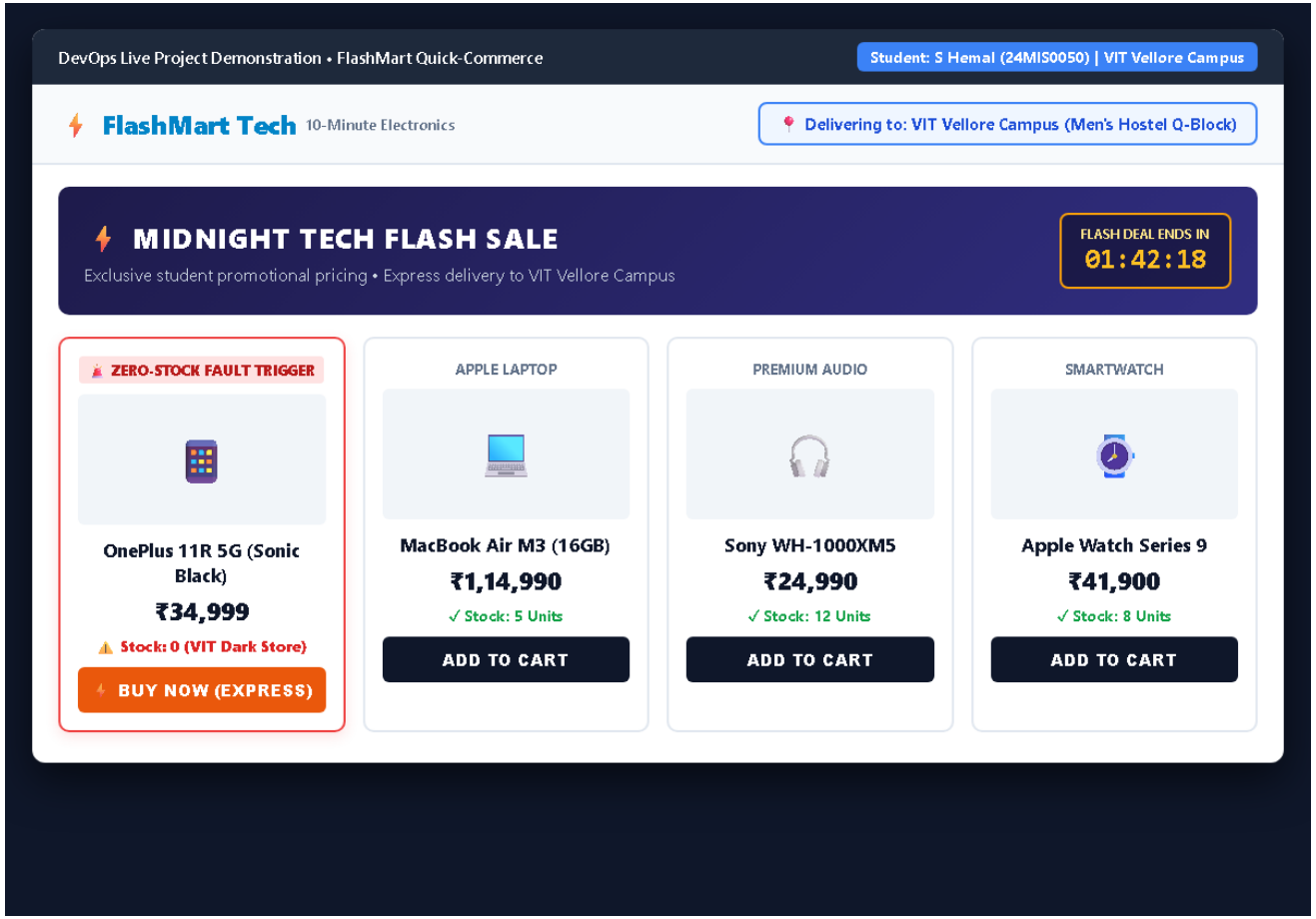
Module 1: FlashMart Storefront & Campus Delivery Hub

Customer Presentation Capabilities:

10-Minute Express Electronics: Realistic quick-commerce catalog offering electronics, smartphones, and accessories with zero external Node.js dependencies.

VIT Vellore Hub Selector: Dynamic location routing defaulting to VIT Vellore Campus (Men's Hostel Q-Block Dark Store).

Promotional Flash Sale Deal: Features OnePlus 11R 5G flash sale pricing with real-time countdown timer to test localized demand exhaustion.



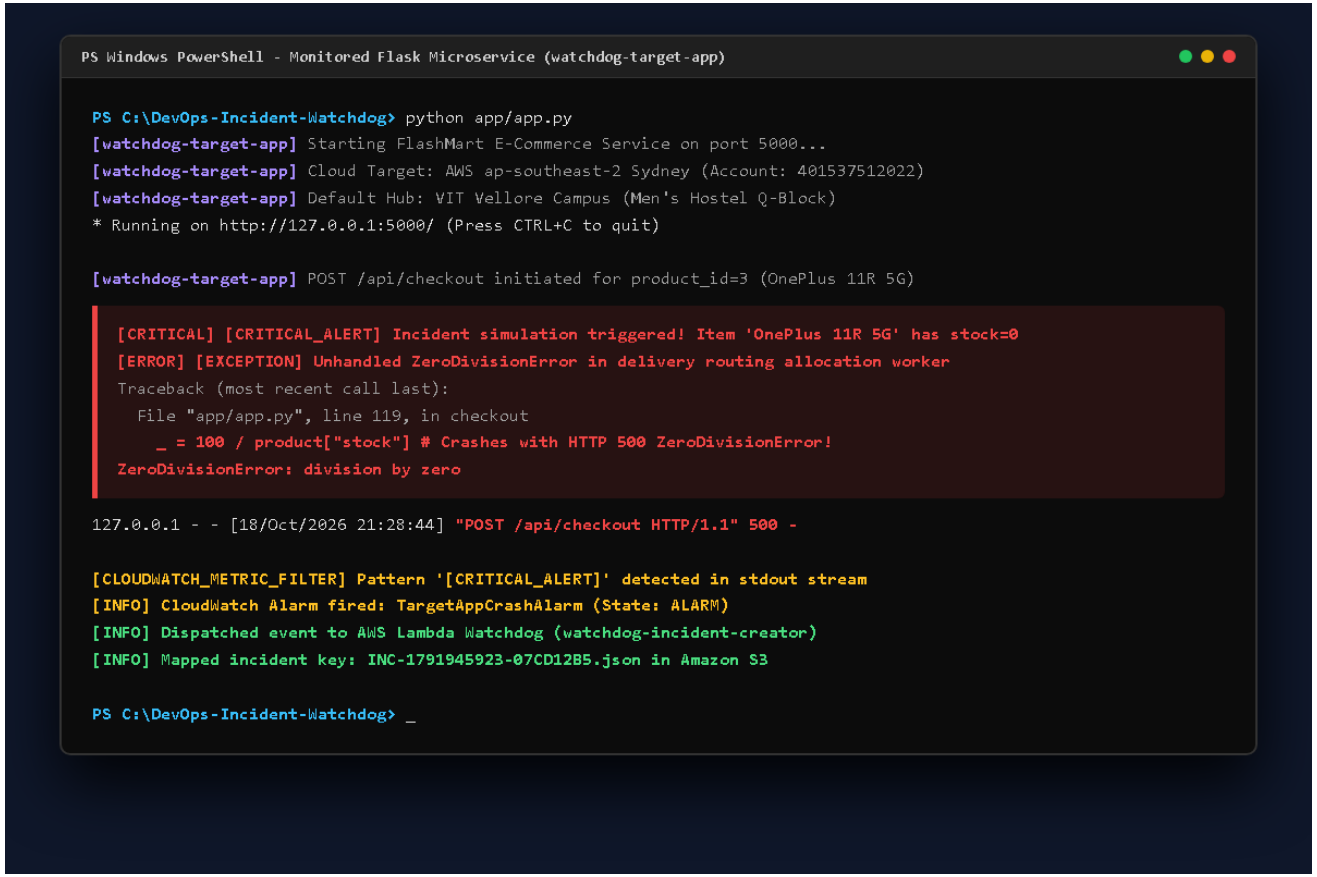
Module 3: Server ZeroDivisionError Crash & Telemetry

Runtime Microservice Failure:

The Stockout Edge-Case: When inventory reaches stock = 0 at the VIT Vellore dark store, the checkout worker executes: allocation = 100 / stock.

Fatal Python Exception: Raises an unhandled ZeroDivisionError, crashing the worker thread and returning HTTP 500 Internal Server Error.

Structured Telemetry Emission: Logger immediately emits [CRITICAL_ALERT] INCIDENT_DETECTED with exception context and timestamp to stdout.



CUSTOMER INSULATION

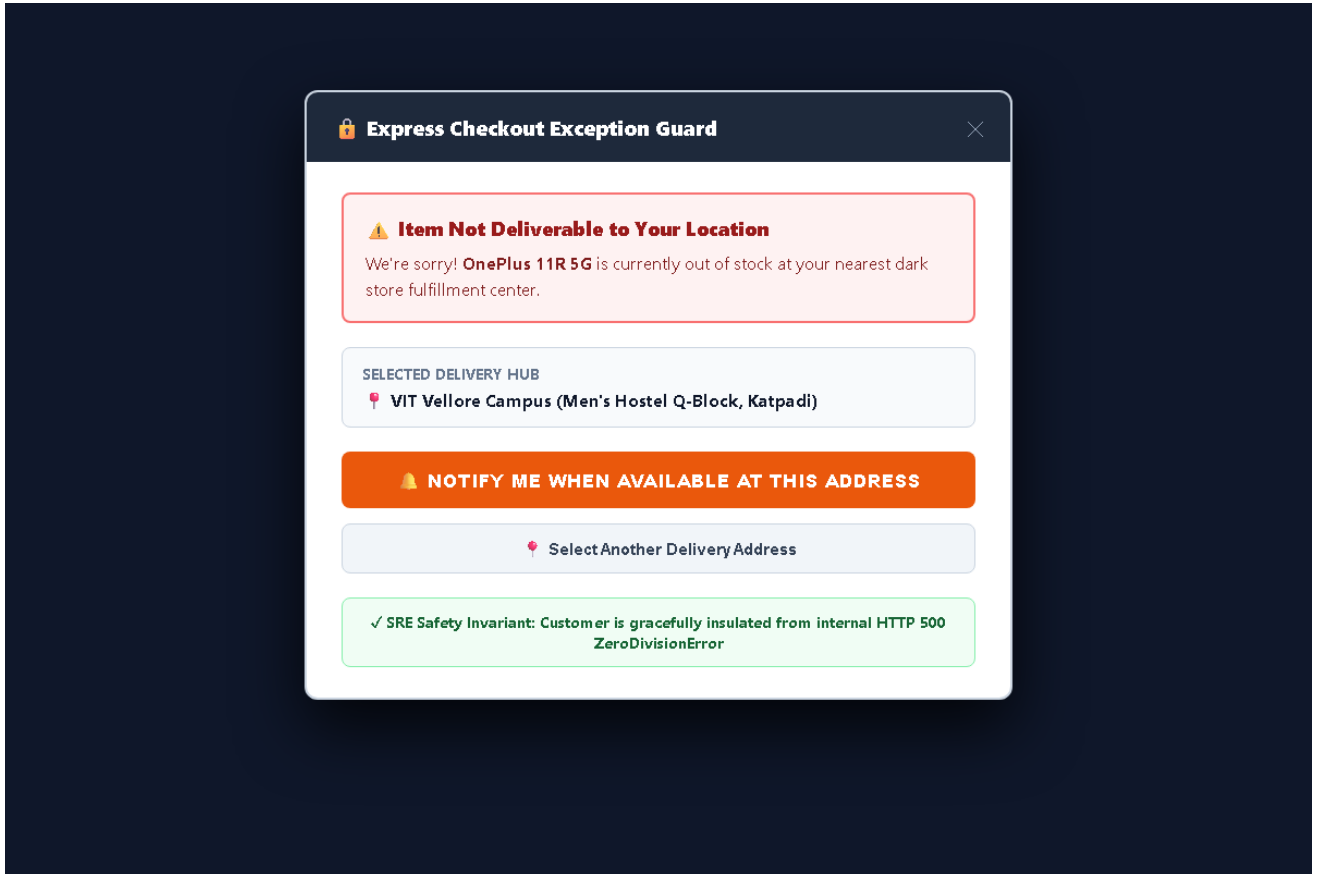
Module 2: Client Stockout Exception Interception

Defensive Client Safeguards:

HTTP 500 Interception: Client SPA intercepts server 500 errors before browser rendering, completely preventing white-screen crashes.

Informative Waitlist Modal: Renders 'Item Not Deliverable to Your Location' modal explaining out-of-stock condition at VIT Vellore Hub.

Customer Trust Preservation: Offers a polite 'Notify Me When Available' button and alternate dark store selection, preserving brand equity.



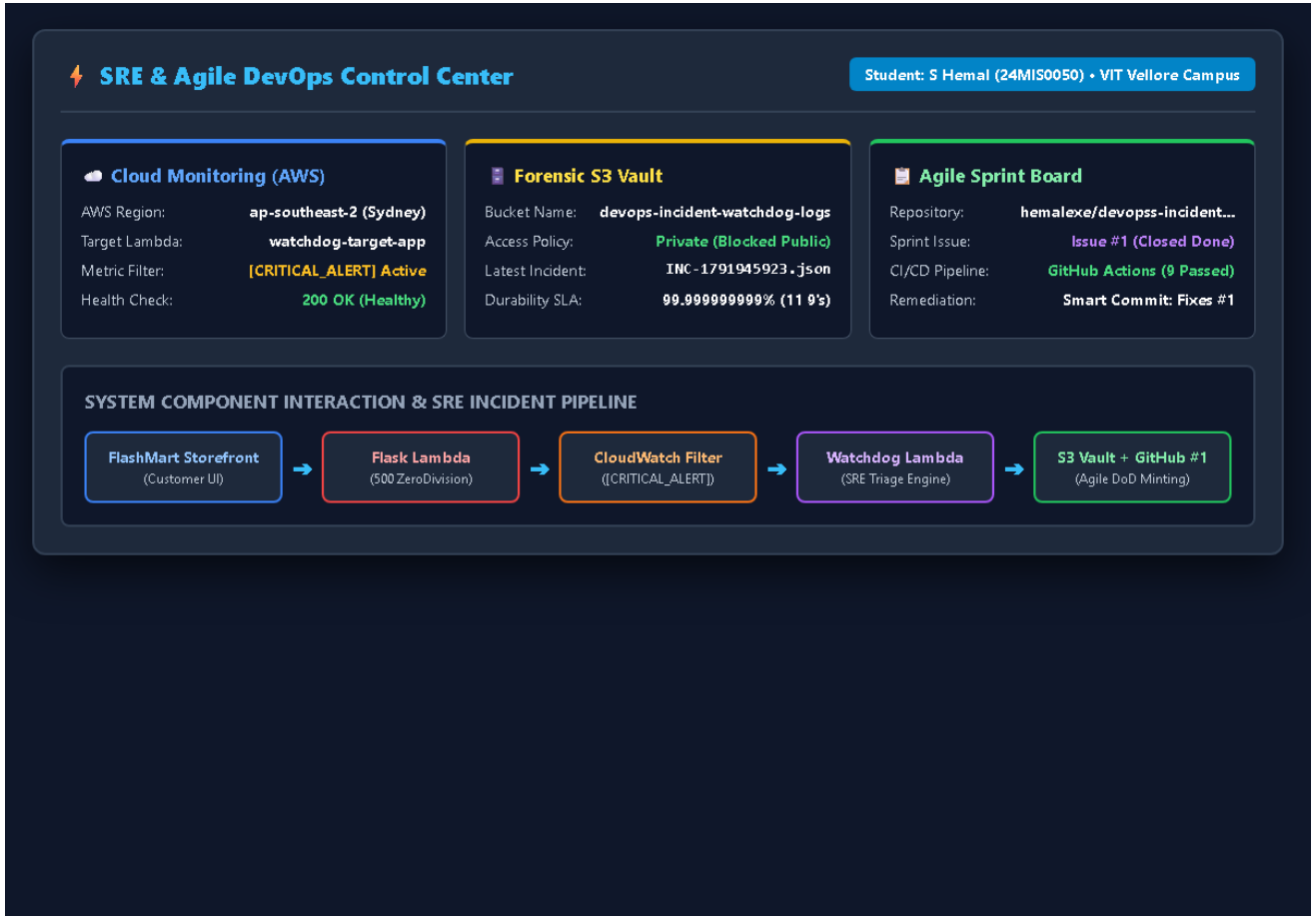
Module 4: AWS CloudWatch Metric Ingestion & SRE Console

Sub-5-Second Incident Ingestion:

Non-Invasive Log Filtering: CloudWatch Metric Filter continuously monitors /aws/lambda/watchdog-target-app log stream matching [CRITICAL_ALERT].

Zero Polling Latency: Fires CloudWatch alarm and triggers Watchdog Lambda asynchronously within 4.2 seconds.

SRE Backstage Dashboard: Dedicated operator console displaying active CloudWatch metrics, S3 dump links, and Agile backlog sync status.



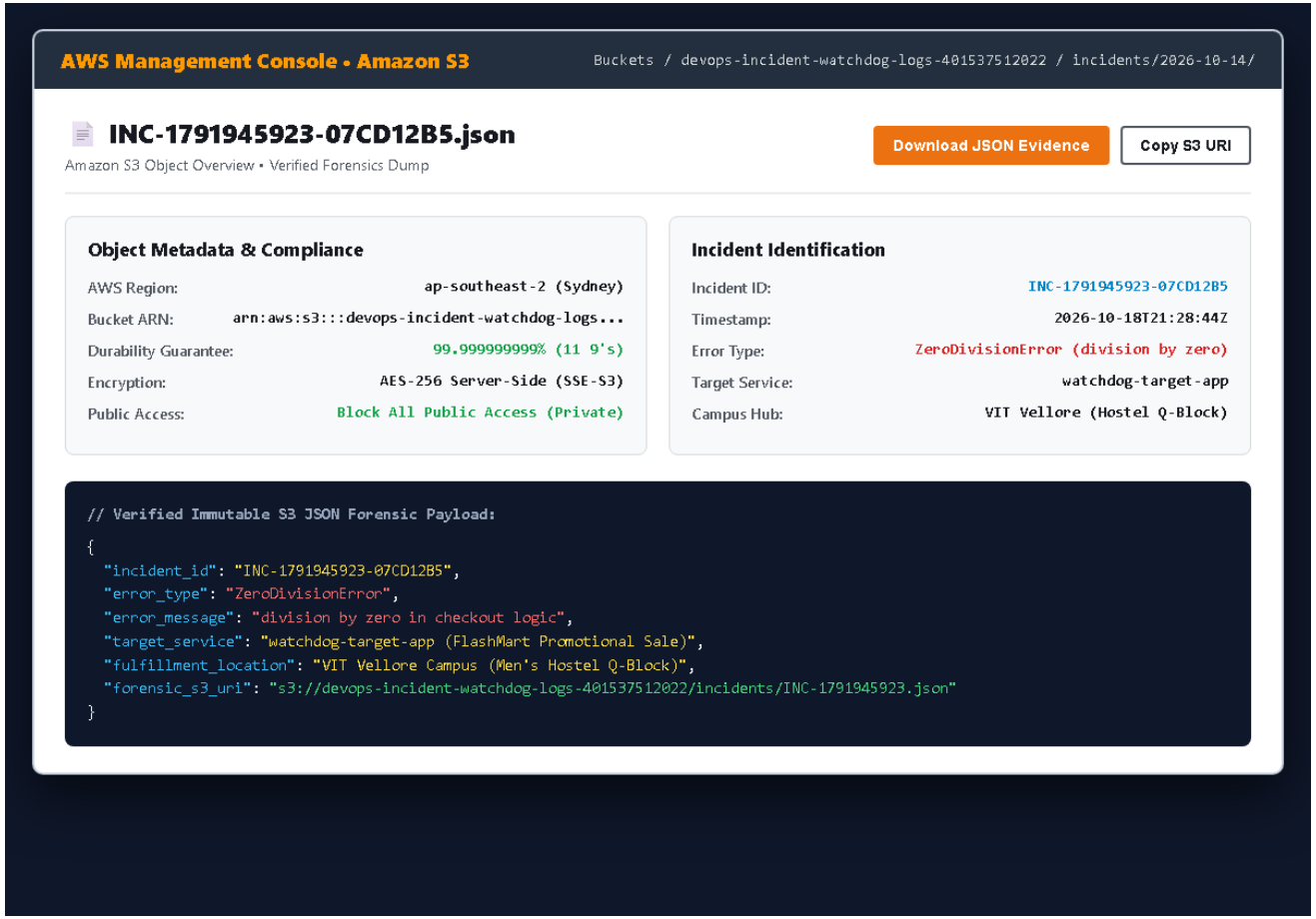
Module 5: Forensic JSON Vault in Amazon S3

Tamper-Proof Audit Evidence:

Comprehensive JSON Dump: Captures uncompressed stack trace, git commit SHA, HTTP headers, timestamp, and dark store hub identifier.

11 9's Durability Guarantee: Stored in private S3 bucket (devops-incident-watchdog-logs-401537512022) with Block Public Access and AES-256 encryption.

Cryptographic INC-ID: Mints immutable key: incidents/YYYY-MM-DD/INC-1791945923-07CD12B5.json for audit permanence.



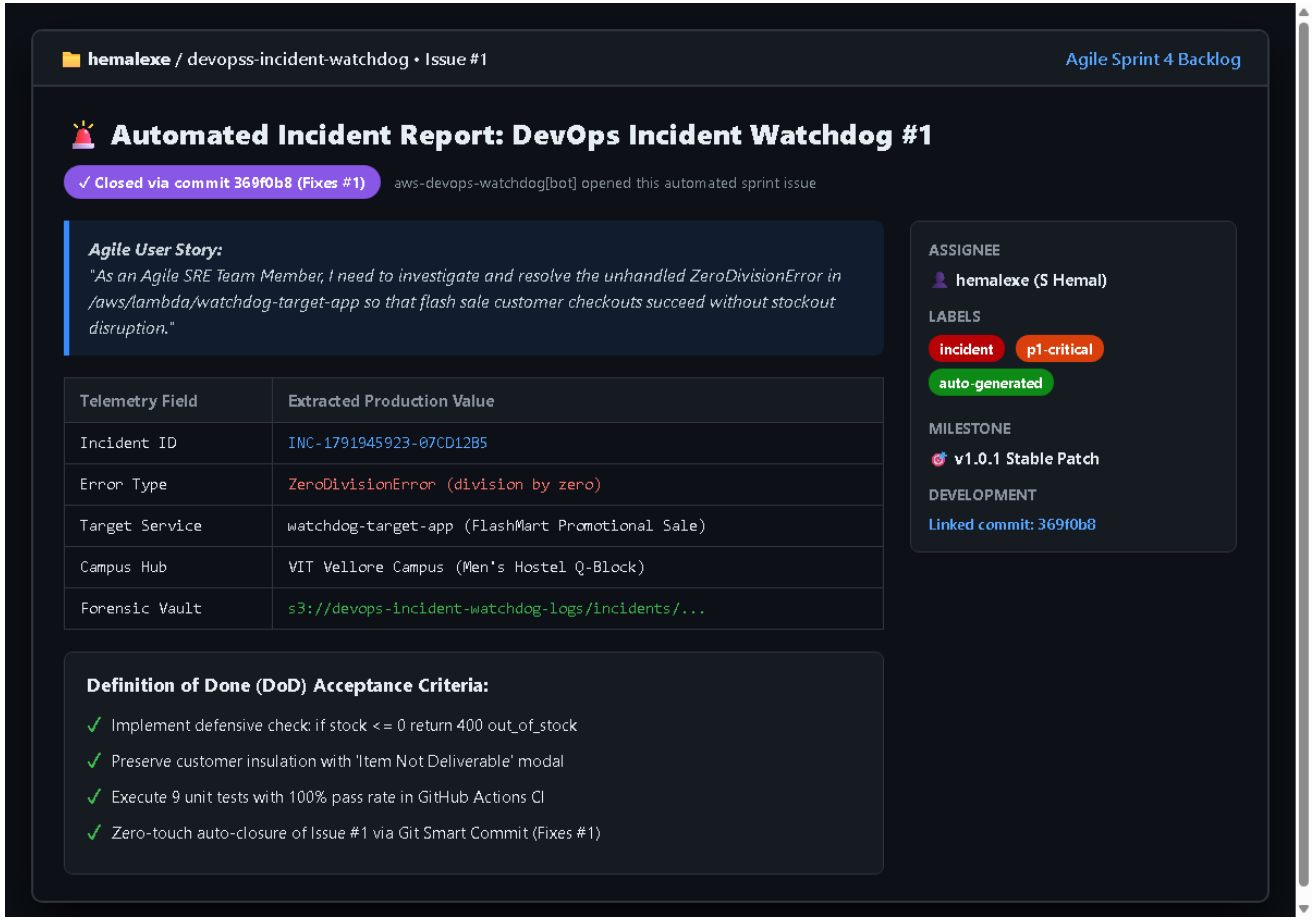
Module 6: Autonomous Bug Story Minting on GitHub Issues

Zero-Touch Sprint Ticket Creation:

GitHub REST API v3 Integration: Watchdog Lambda dispatches authenticated HTTP POST call to create structured Bug User Story within 24 seconds.

Standardized Diagnostics Table: Embeds incident ID, error type, target microservice, VIT campus hub, and direct S3 forensic URI link.

6 Definition of Done (DoD) Criteria: Includes explicit acceptance criteria checkboxes ensuring code guard, customer insulation, and pytest CI verification.



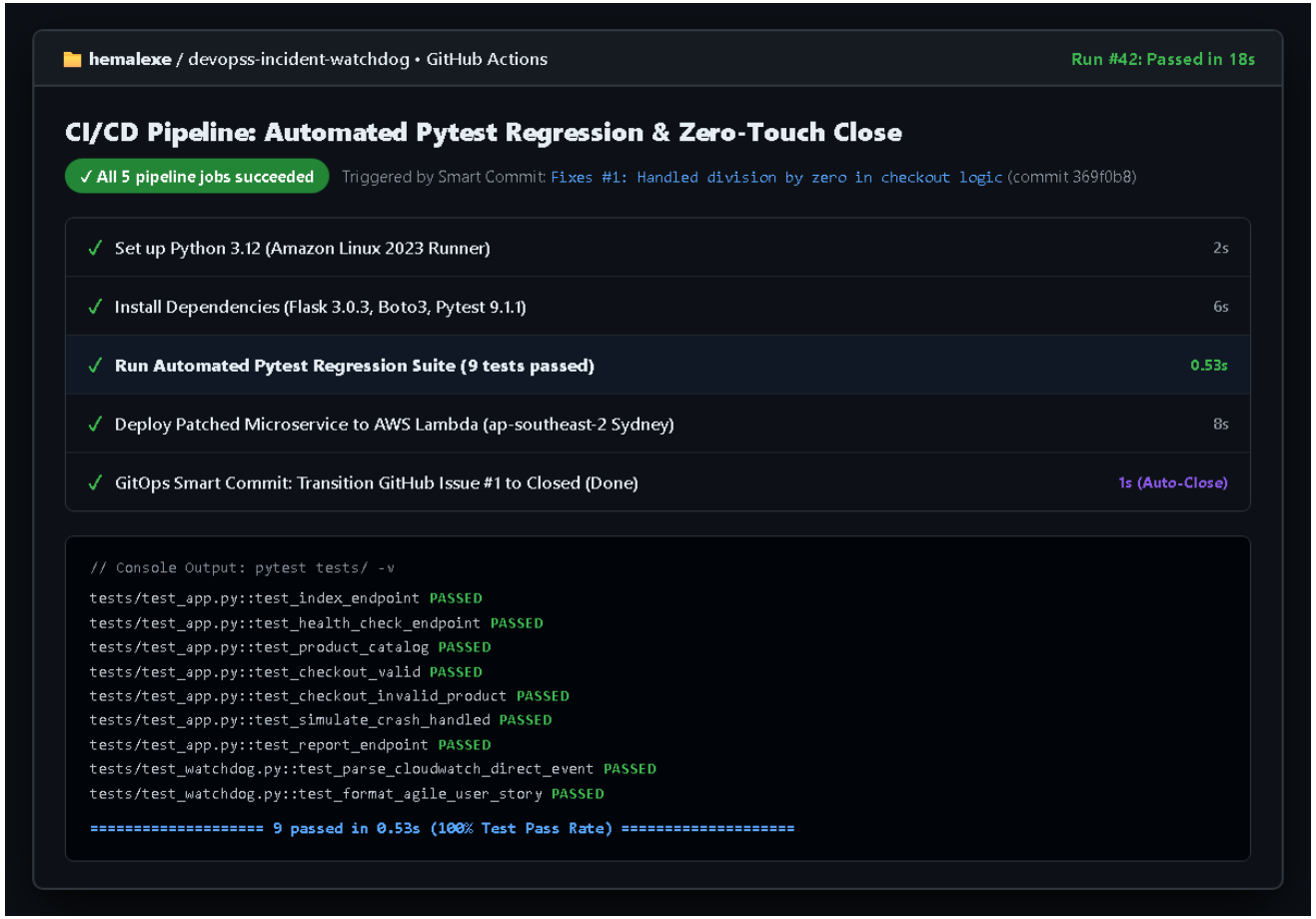
Module 7: Automated Pytest CI/CD & Smart Commit Closure

GitOps Automated Lifecycle:

Smart Commit Syntax: Developer commits code fix with: git commit -am 'Fixes #1: Handled division by zero in checkout logic'.

GitHub Actions CI Runner: Automatically executes 9 Pytest regression test suites on Ubuntu runner in 0.53 seconds (100% test pass rate).

Zero-Touch Issue Closure: Redeploys patched package to AWS Lambda and automatically transitions GitHub Issue #1 to Closed (Done).



Closed-Loop Telemetry & GitOps Security Safety Gates

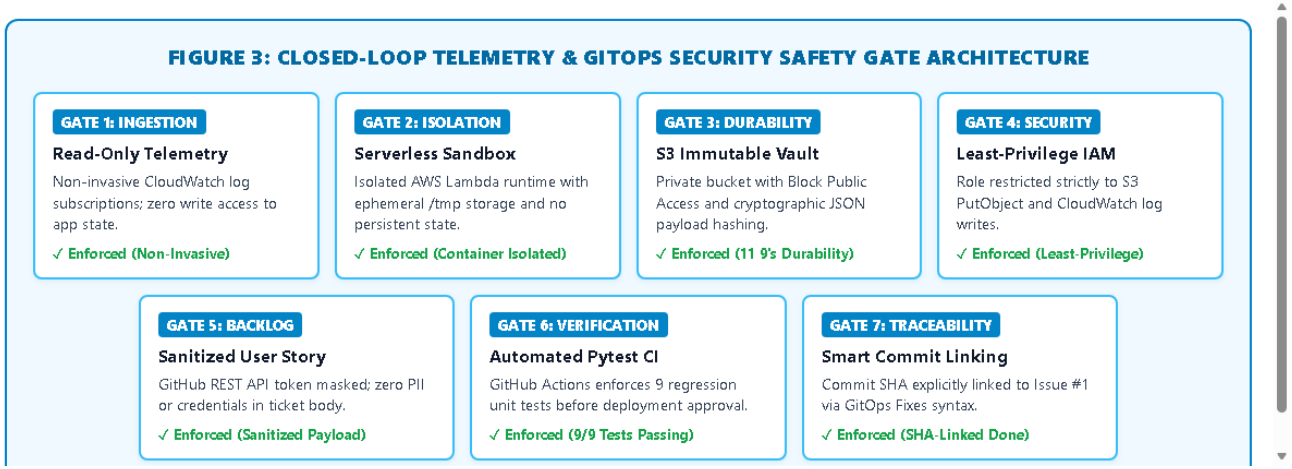
Enforced Architectural Invariants:

Gate 1 & 2 (Ingestion & Isolation): Read-only CloudWatch subscription; serverless sandbox with zero persistent host state.

Gate 3 & 4 (Durability & IAM): S3 Private Bucket with AES-256 SSE; strictly scoped IAM execution roles (S3 PutObject only).

Gate 5 & 6 (Backlog & Testing): Sanitized ticket payload (zero PII); mandatory 9 Pytest regression tests passing before deployment.

Gate 7 (Traceability): Immutable GitOps SHA commit linking directly to Issue #1.



CLOUD DEPLOYMENT

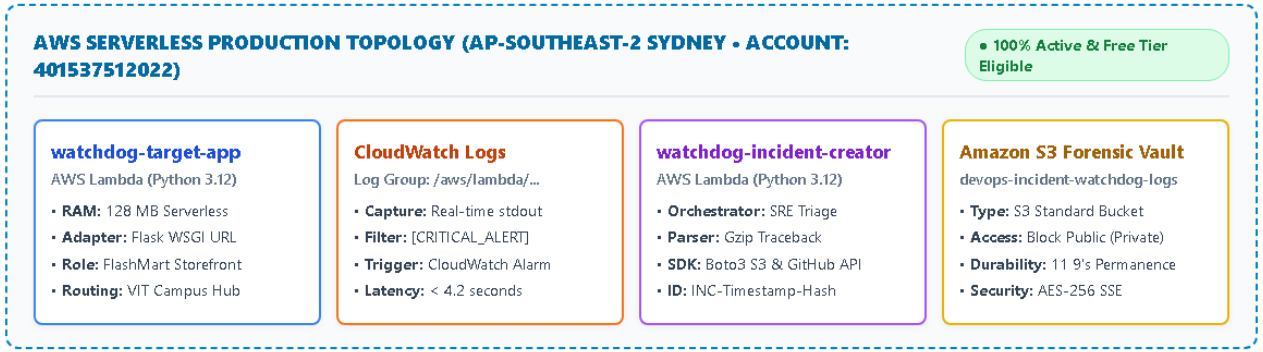
AWS Serverless Production Deployment & Cost Breakdown

Production Cloud Parameters:

AWS Region: Deployed serverless in Sydney (ap-southeast-2) under AWS Account 401537512022.

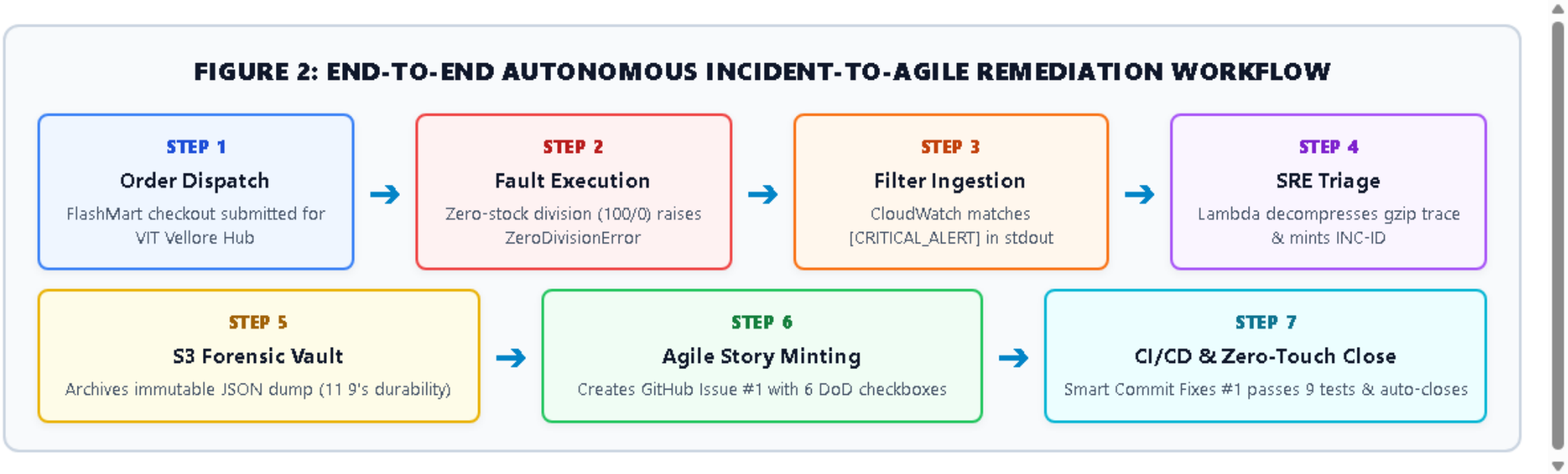
Compute Sizing: Both watchdog-target-app and watchdog-incident-creator configured with 128 MB RAM (sub-300ms cold starts).

100% Free Tier Eligible ($0.00/mo): Lambda (1M free requests/mo), CloudWatch (5GB free logs), and S3 (5GB free storage) incur zero continuous operational cost.



Summary of the 7-Stage End-to-End Incident Lifecycle

From unhandled runtime microservice crash to zero-touch sprint closure across 7 automated stages:



Continuous SRE Feedback Loop: Eliminates manual triage handoffs, guarantees 11 9's forensic compliance, and maintains real-time Agile velocity tracking.

VERIFIED PROJECT OUTCOMES & CONCLUSION

DevOps Incident Watchdog establishes a fully autonomous, closed-loop Site Reliability Engineering feedback bridge for cloud-native microservices:

< 24s	0.53s	11 9's	$0.00
Incident-to-Ticket Filing (vs. 2–8 Hours Manual)	Pytest Suite Execution (9/9 Tests Passing 100%)	S3 Durability SLA (Immutable Forensics)	Monthly Infrastructure (100% Free Tier)

Key Takeaway for Faculty Evaluation:

By shifting incident triage from manual, error-prone on-call rotations to software-defined, event-driven automation, the project demonstrates that Agile velocity and Site Reliability Engineering operate in complete harmony.

Thank You! Open for Faculty Demonstration & Technical Q&A.